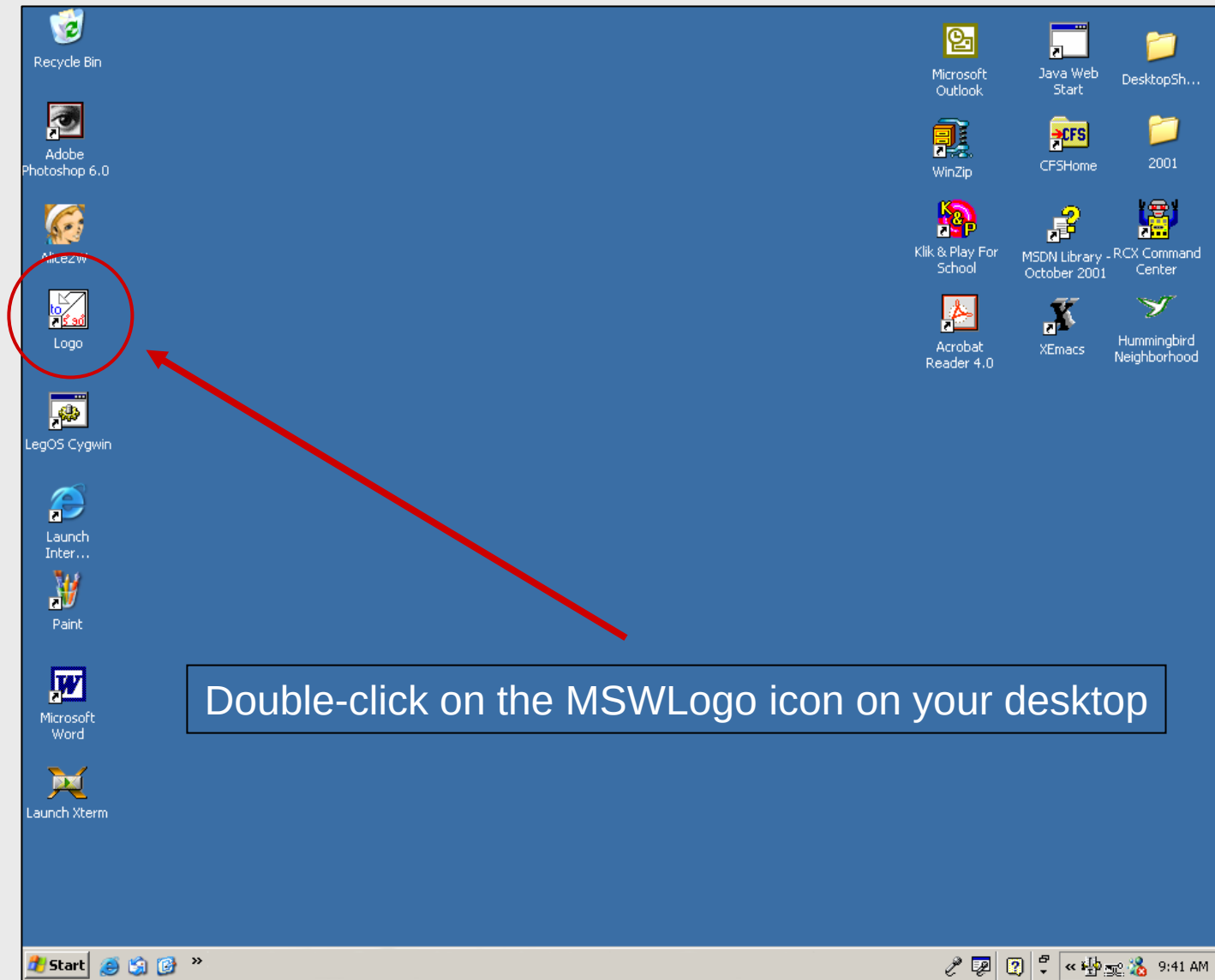


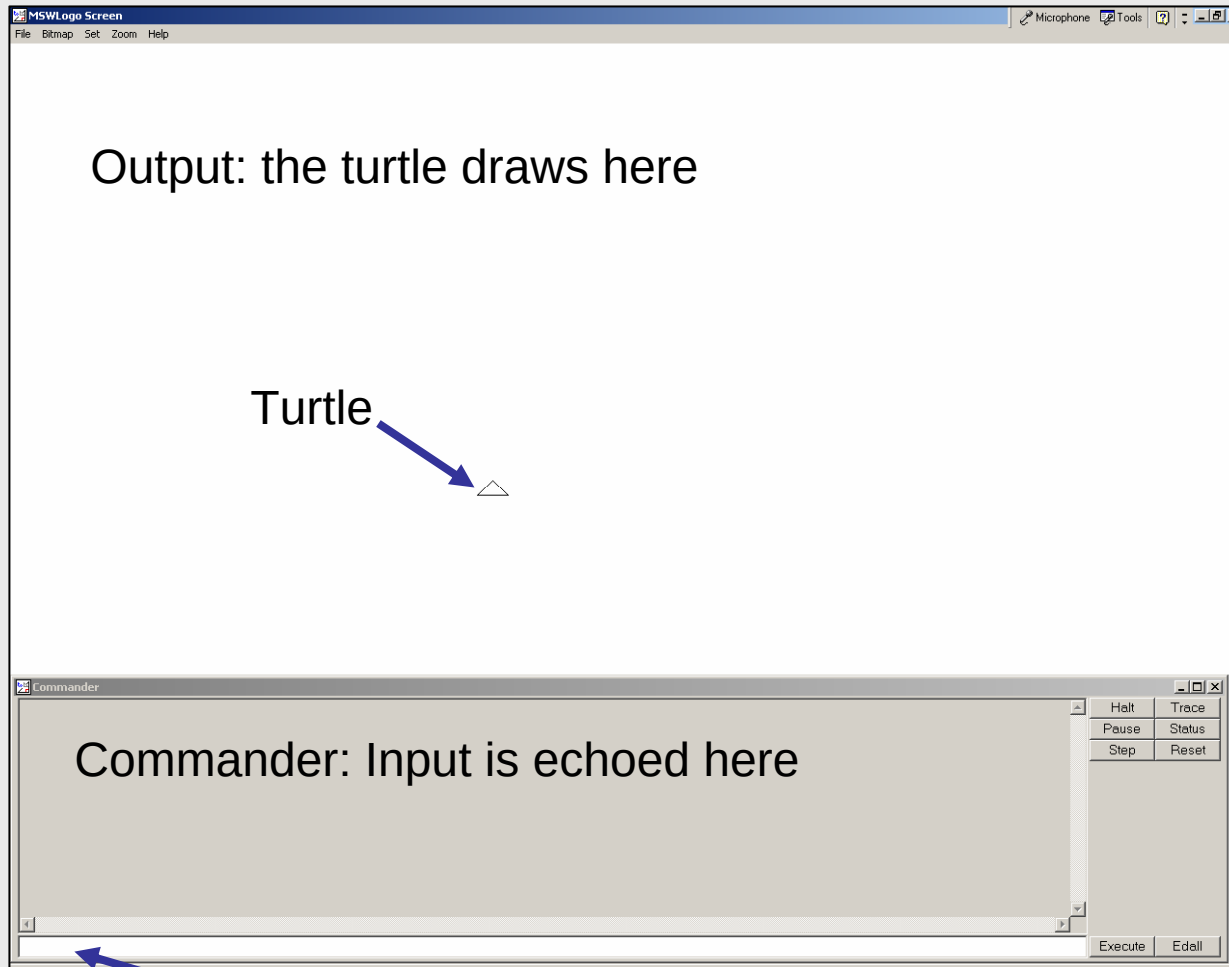
LOGO



Starting Logo



Understanding the Logo Screen

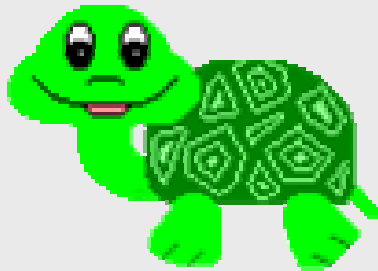


Input box: type your code commands in here

A Little Logo History

What is LOGO?

- A programming language developed at the MIT Artificial Intelligence Lab
- It's easy to learn, but once you know the basics you can do extremely complicated things
- Logo code has been used in telecommunications, multimedia software and robotics



What is the TURTLE?

- Originally, the turtle was a robotic creature that sat on the floor and was directed to move around by a user typing commands into a computer
- Today, it is an icon on the Logo screen (in our version of Logo the turtle is represented by a triangle)



Basic Logo Commands

FORWARD This command is followed by a **number**. The larger the number the farther the turtle will go. If you use a large enough number, the turtle will fall off the screen and wrap around to the other side.

FORWARD 40

BACK Follow this command with a **number**, as you did for FORWARD. This time the turtle will move backwards.

BACK 33

RIGHT This command is also followed by a **number**. The number represents how many degrees you want the turtle to turn. The example below will turn the turtle to the right 90 degrees.

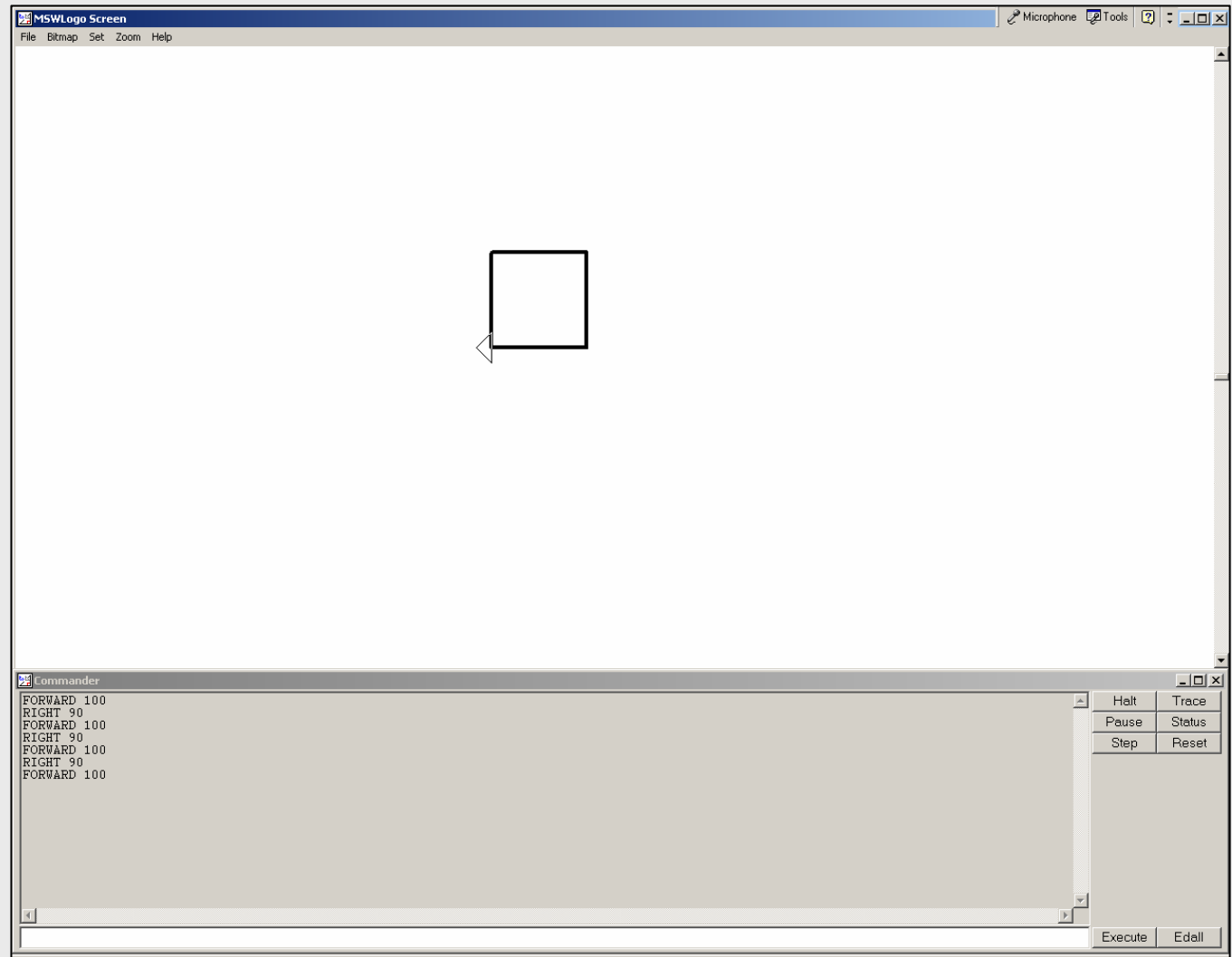
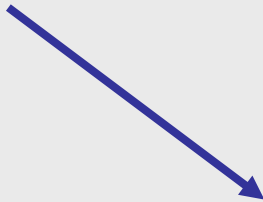
RIGHT 90

LEFT This command works exactly like the RIGHT command, only it turns the turtle to the left.



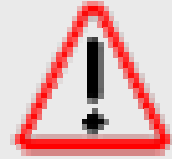
To draw a SQUARE:

FORWARD 100
RIGHT 90
FORWARD 100
RIGHT 90
FORWARD 100
RIGHT 90
FORWARD 100





IMPORTANT



When you are typing commands into the input window, you might notice that sometimes there are messages in the commander window that you didn't type in.

These are called **ERROR MESSAGES**.

When you give Logo input that it doesn't understand, this is how it tells you about what it didn't get. Here are a few examples of Logo **ERROR MESSAGES**:

Your Input: Forward 10 20

Logo's Error Message: "You don't say what to do with 20"

Your Input: Back

Logo's Error Message: "not enough inputs to back"

Other Logo Commands

PENUP This command will cause the turtle to pick up its “pen” so you can move the turtle without drawing a line.

PENDOWN This command is used to put the “pen” back down so you can draw again.

SETPENCOLOR Follow this command with a **number**. You can change the color of the turtle’s “pen” depending on the number you choose. The example below will make your turtle draw in **hot pink**!

SETPENCOLOR 5

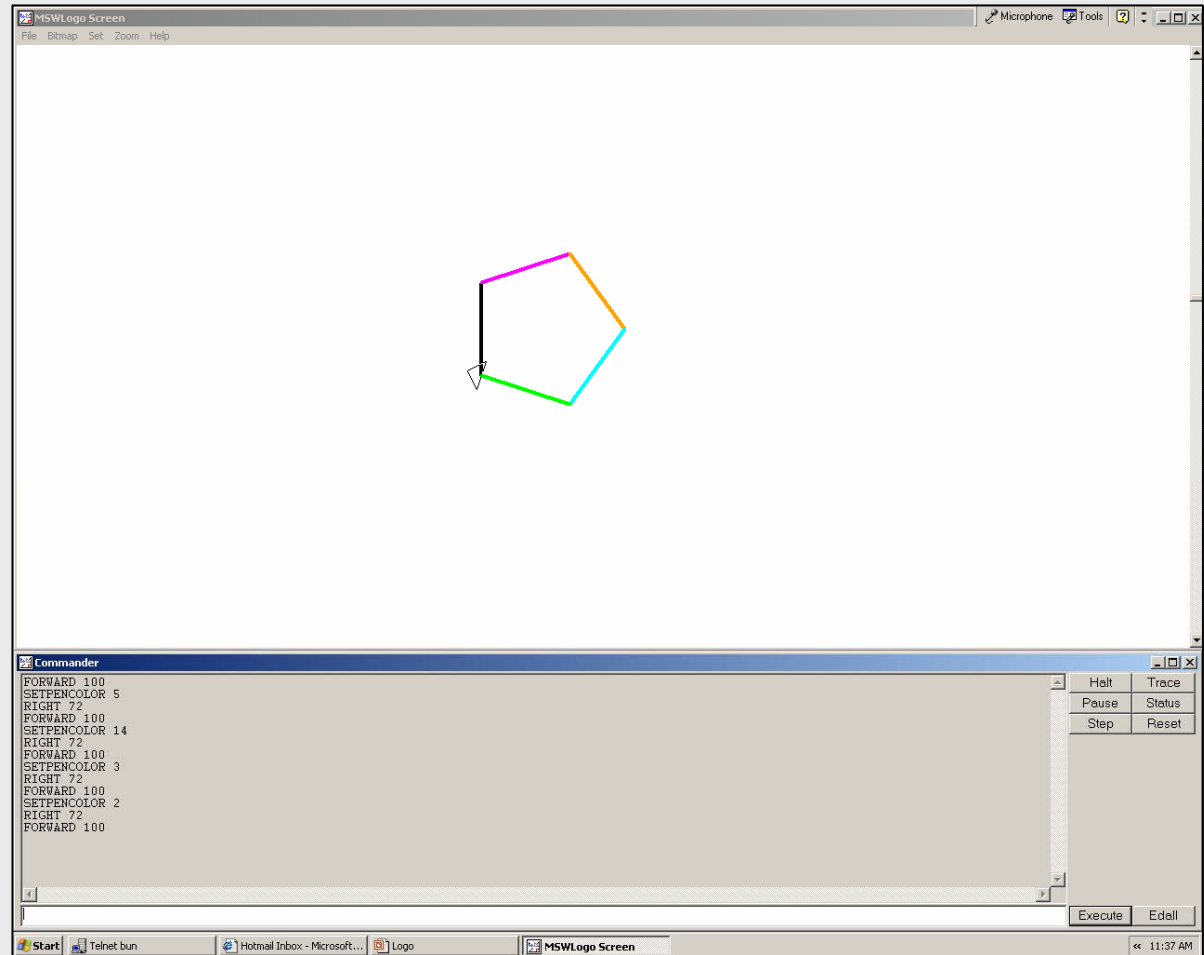
CLEAN This command will erase the canvas (another name for the screen)

HOME This command will move the turtle back to the center of the canvas



An Example using SETPENCOLOR:

```
FORWARD 100  
SETPENCOLOR 5  
RIGHT 72  
FORWARD 100  
SETPENCOLOR 14  
RIGHT 72  
FORWARD 100  
SETPENCOLOR 3  
RIGHT 72  
FORWARD 100  
SETPENCOLOR 2  
RIGHT 72  
FORWARD 100
```



Because the turtle turned right 72 degrees 5 times, it made a **PENTAGON**

The REPEAT Command

You can get your turtle to do one (or several) things repeatedly, without typing them again and again using the REPEAT command.

This example will move the turtle forward 10 spaces, 4 times. In all, the turtle will move forward 40 spaces.

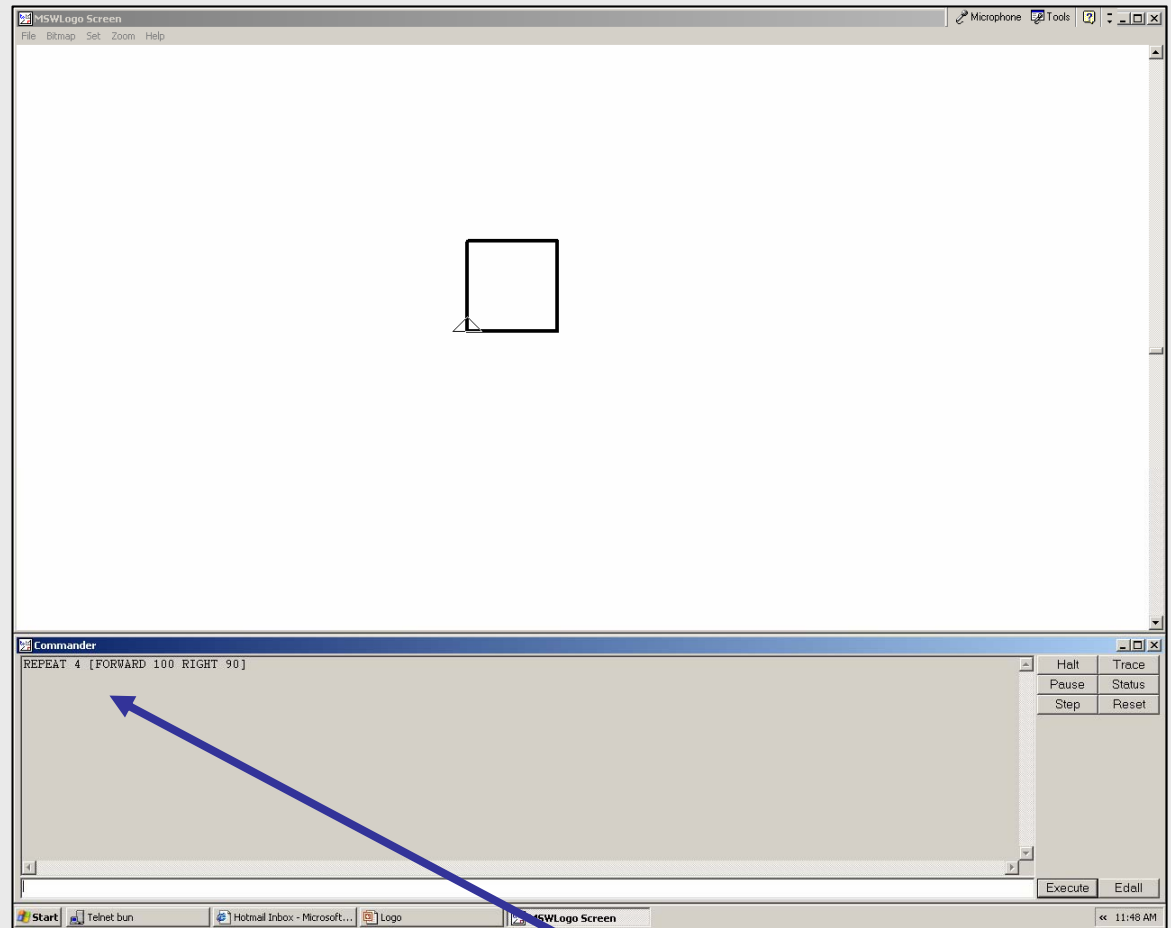
```
REPEAT 4 [FORWARD 10]
```

Can you think of a way the REPEAT command would be useful for drawing a triangle, square, or pentagon?



Let's try drawing a SQUARE again: (using REPEAT this time)

Wow! That
sure is a lot
less typing!



This time, the only command was: **REPEAT 4 [FORWARD 100 RIGHT 90]**

Try These Questions:

1. Can you make a square yourself? Can you change its size? How about its color?
2. Can you figure out what this command will draw before you try it?

`REPEAT 3 [FORWARD 100 LEFT 120]`

Now type it in and see if you were correct.

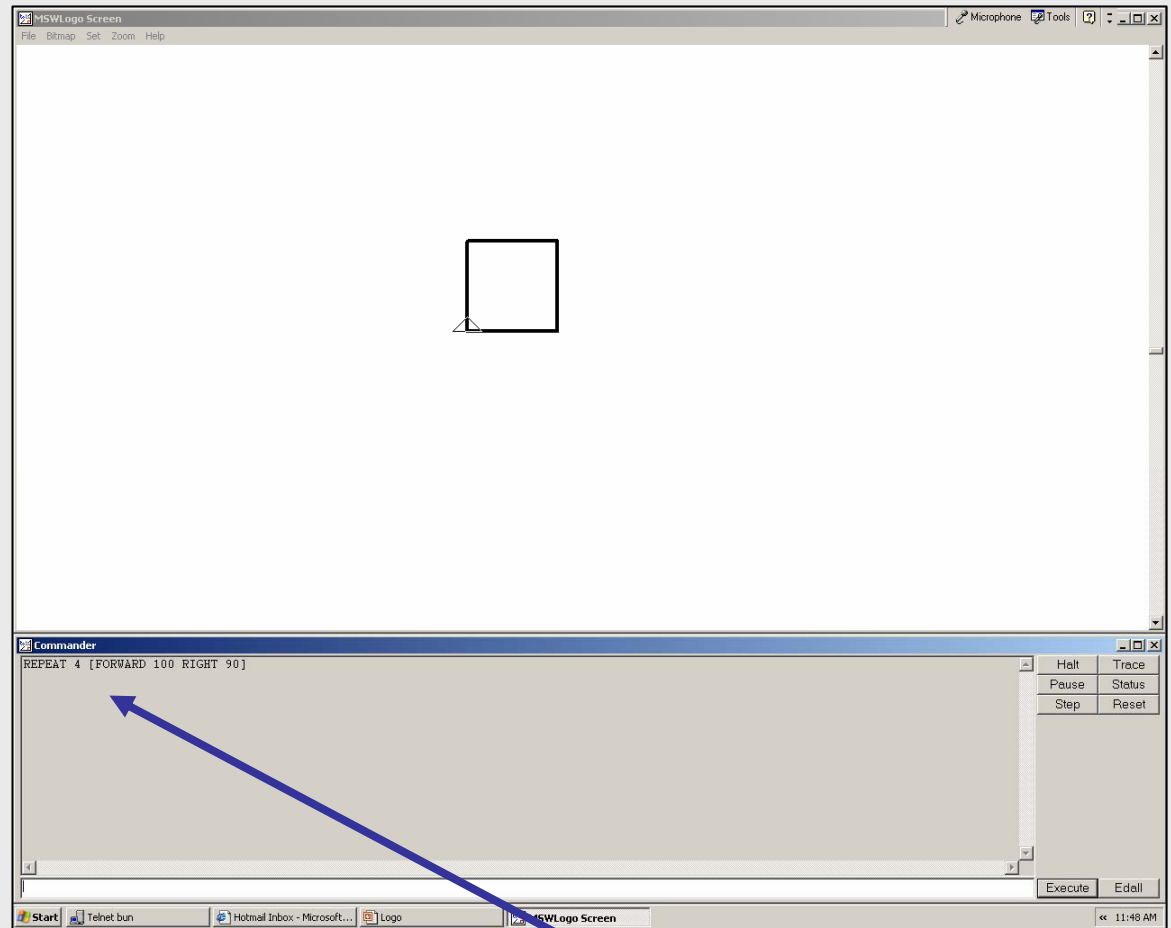
3. Here's a difficult one: Try to make a circle using the REPEAT command.



Answer 1:

Turtle draws a SQUARE

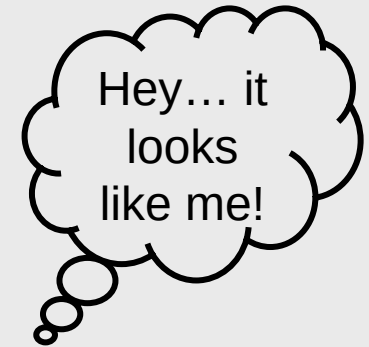
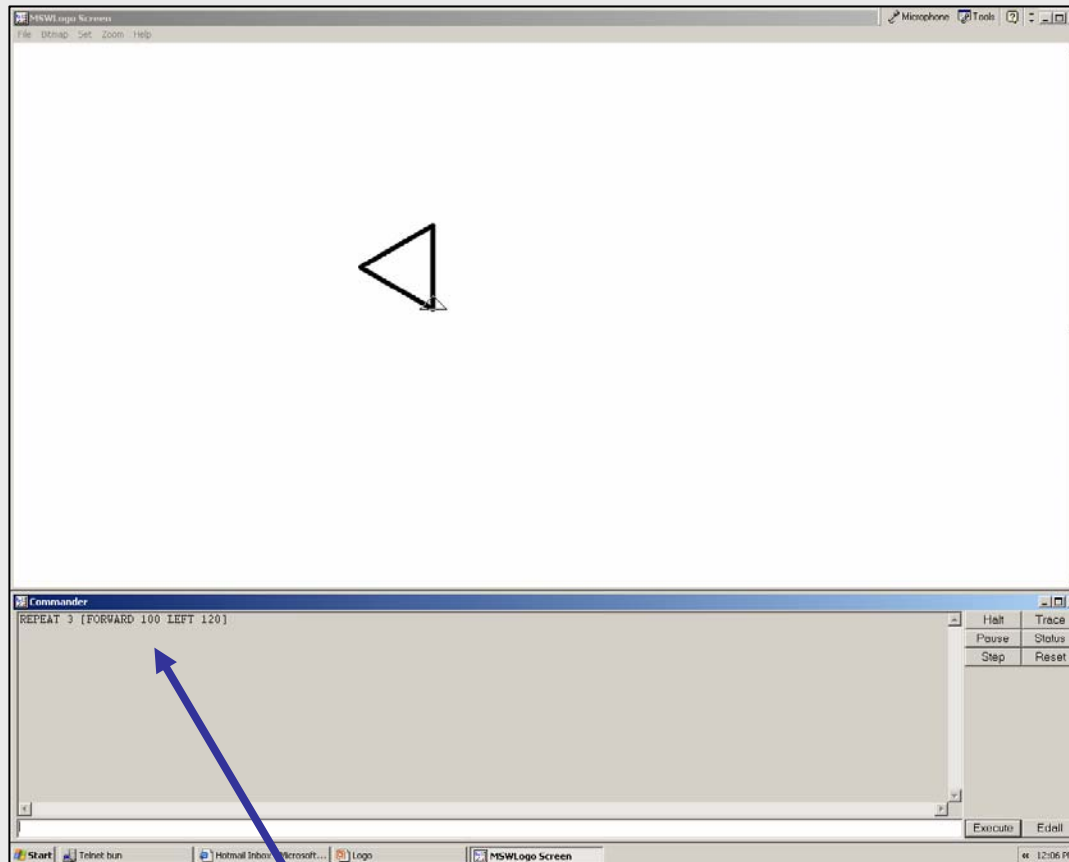
Here's our
square again:



The command was: **REPEAT 4 [FORWARD 100 RIGHT 90]**

Answer 2:

Turtle draws a TRIANGLE

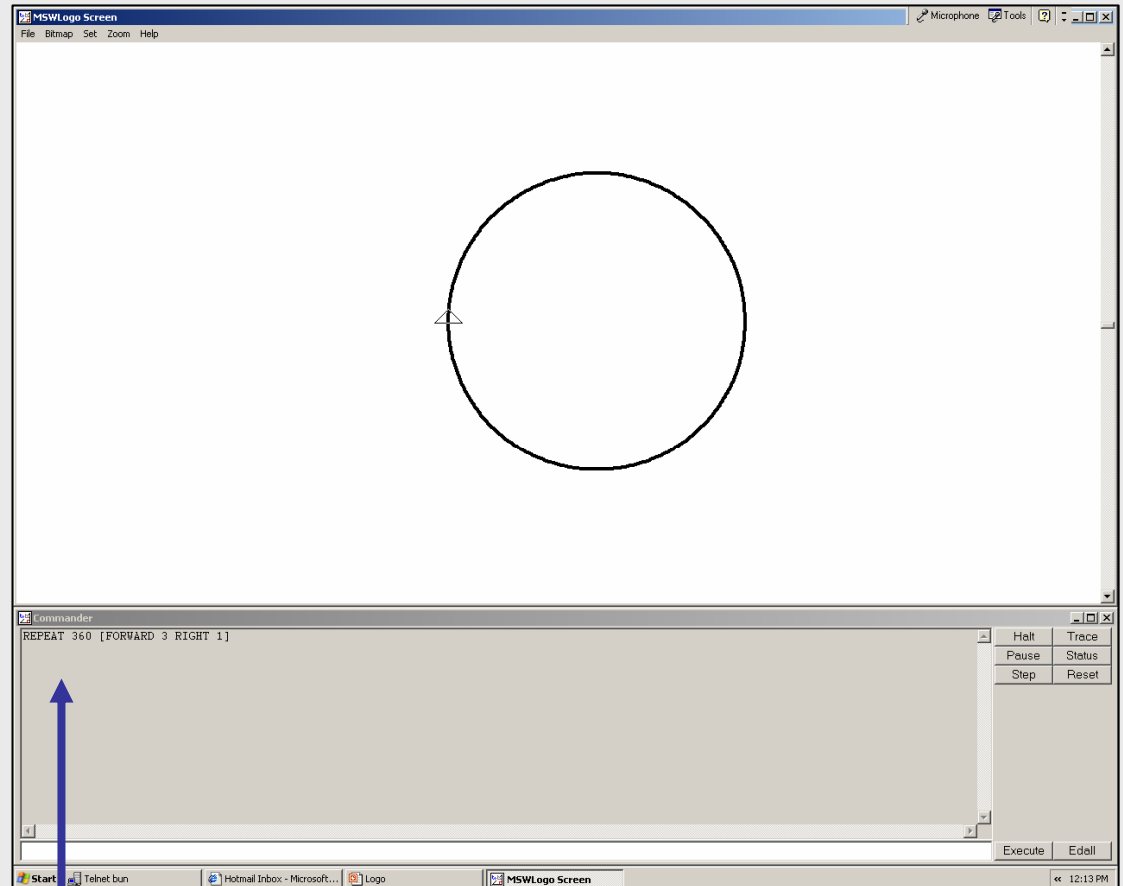


REPEAT 3 [FORWARD 100 LEFT 120]

Answer 3:

Here's one way to make a CIRCLE

Does your circle
look like this
one? What did
you do
differently?



REPEAT 360 [FORWARD 1 RIGHT 1]

Programs in LOGO

You can teach Logo how to draw a square or a circle by making it a **PROGRAM**.

Every time you want to write a program it must be in this form:

```
TO <name of your program>  
<command 1>  
<command 2>  
...  
END
```

For example:

```
TO square  
REPEAT 4 [FORWARD 100 RIGHT 90]  
END
```

name of the program

the commands

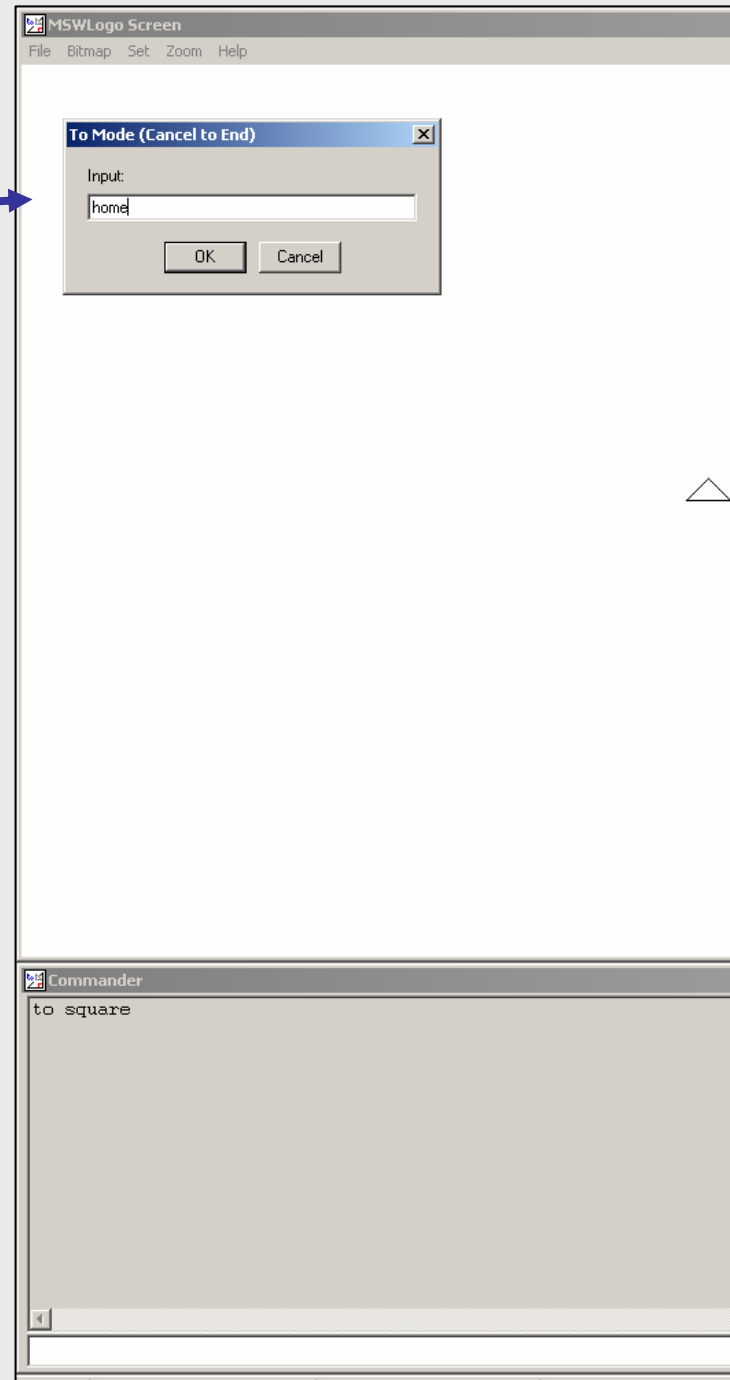


NOTE:

a window will pop up
after you type

TO <name of program>

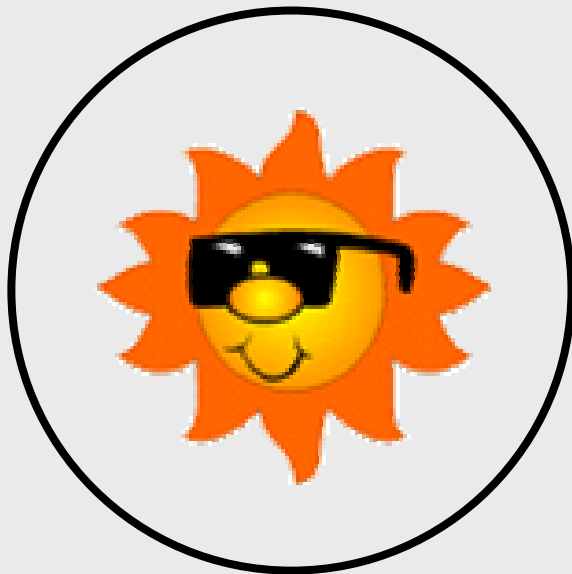
and you can enter your
program line by line
and either type **end** or
hit **Cancel** to finish!



Your First LOGO Program

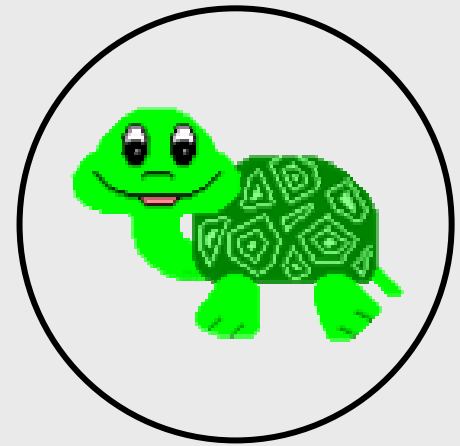
Try making a program that will draw a circle.

Remember to use TO and END. Don't forget to give your program a name.



After writing the program, test it out by typing in the program name.

CIRCLE



The program:

```
TO CIRCLE  
REPEAT 360 [FORWARD 2 RIGHT 1]  
END
```

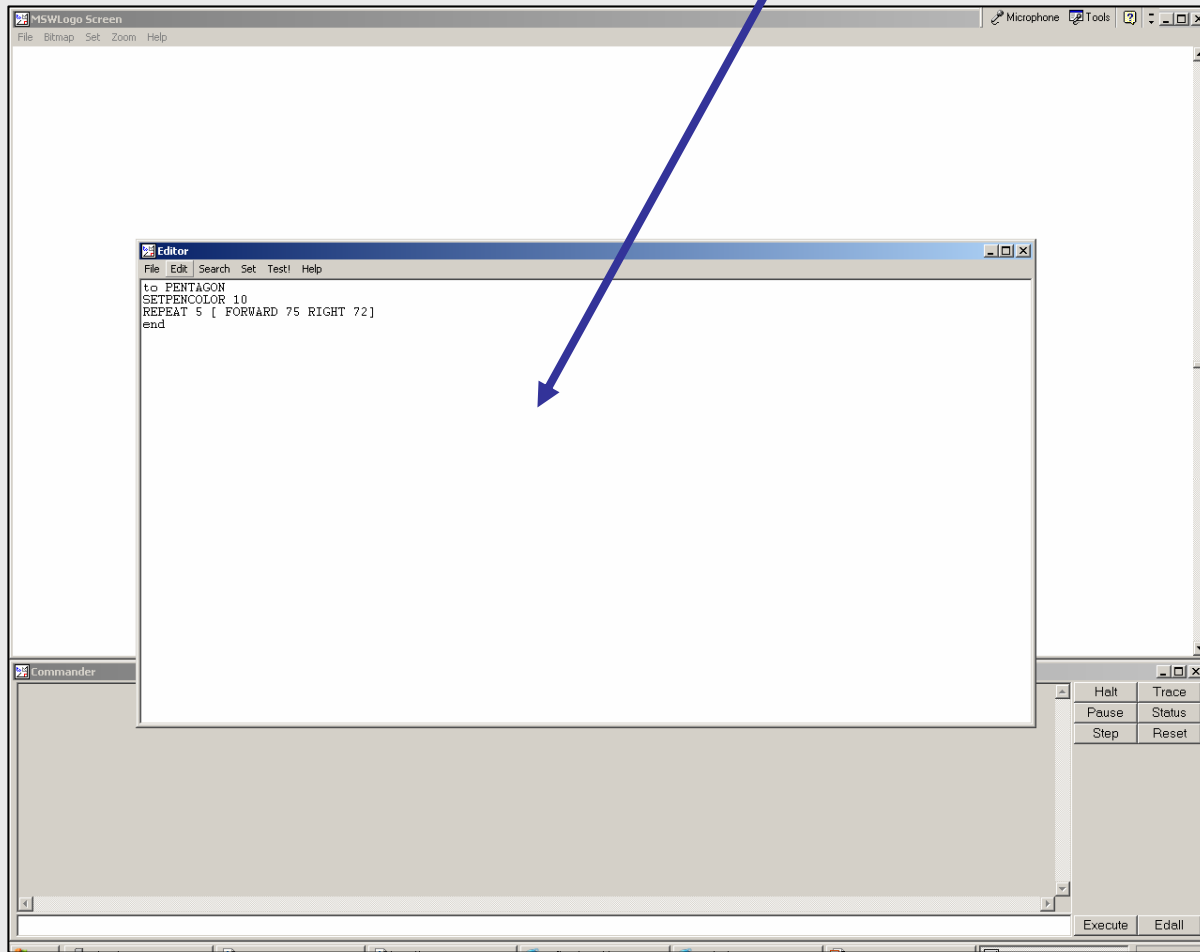
Test it out:

```
CIRCLE
```

If you want to add more programs, or change a program you already wrote, click on the “EDALL” button in the bottom right corner of the Logo window.



The "EDALL" Window



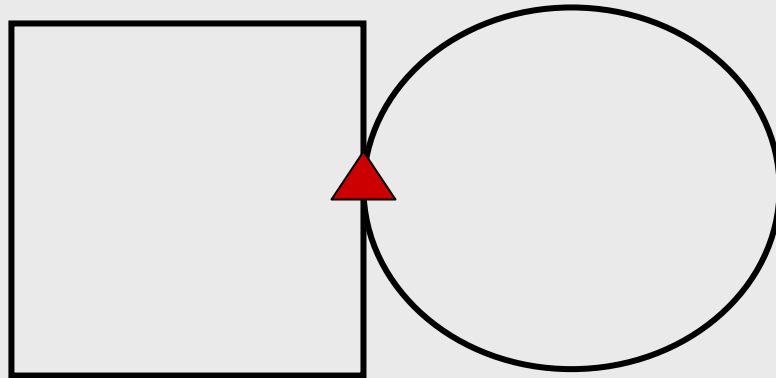
This program will draw a **purple** pentagon:

```
TO PENTAGON  
SETPENCOLOR 10  
REPEAT 5  
  [FORWARD 75  
   RIGHT 72]  
END
```

Try writing several programs in the "EDALL" window.

A Slightly More Complicated Program

How would you write the program `CIRCLE_AND_SQUARE` to make a drawing that looks like this?



The red triangle is the position of the turtle at the end of the program.



CIRCLE_AND_SQUARE

Here's one way to write a program that draws the circle and square as it was on the previous slide:

```
TO CIRCLE_AND_SQUARE  
HOME  
CLEAN  
REPEAT 360 [FORWARD 1 RIGHT 1]  
FORWARD 50  
REPEAT 3 [LEFT 90 FORWARD 100]  
LEFT 90  
FORWARD 50  
END
```



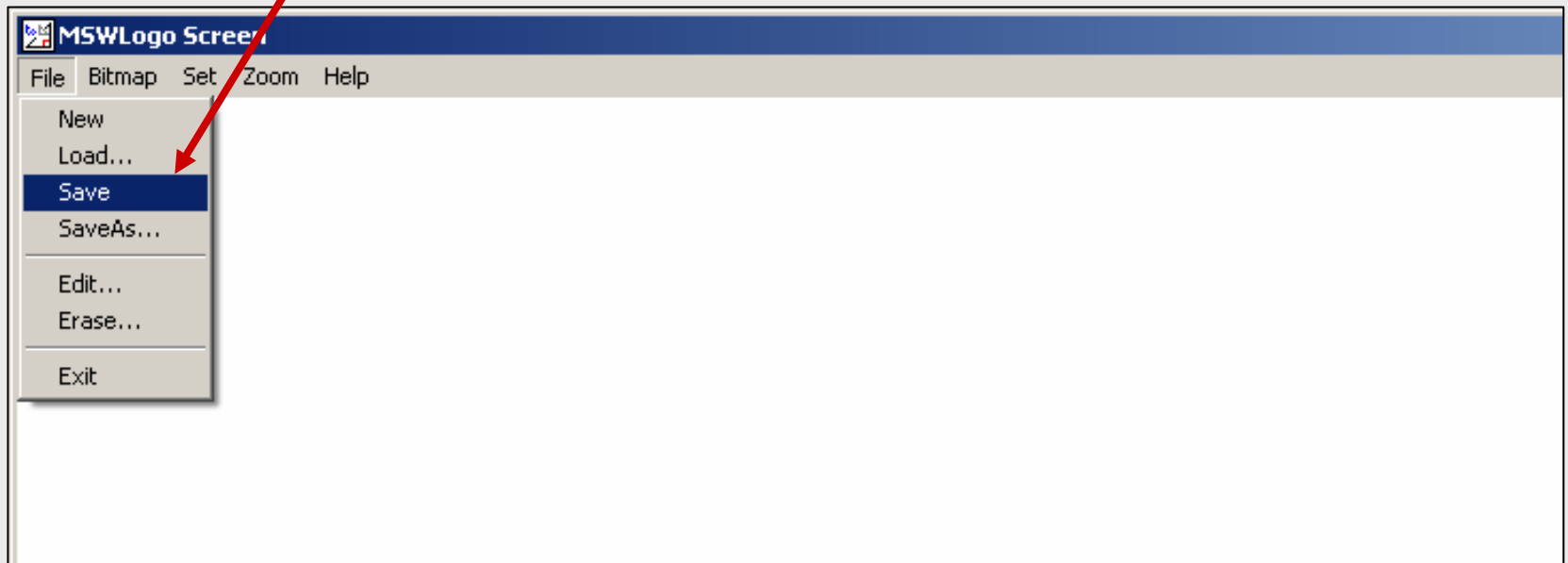
Can you think of how your **CIRCLE** program could be useful in the **CIRCLE_AND_SQUARE** program?
Change **CIRCLE_AND_SQUARE** so it uses **CIRCLE**.



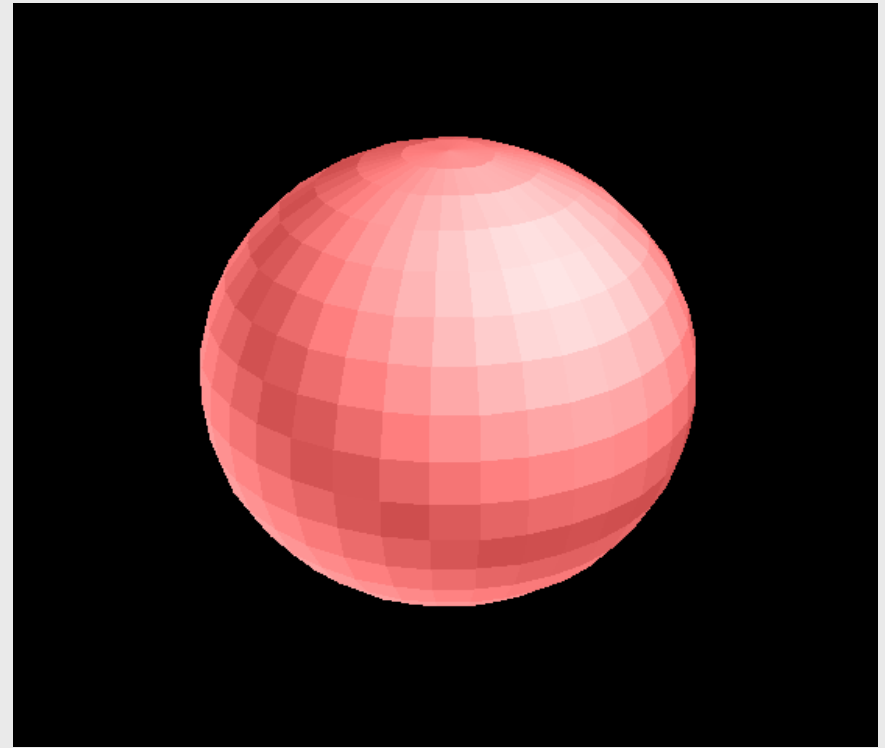
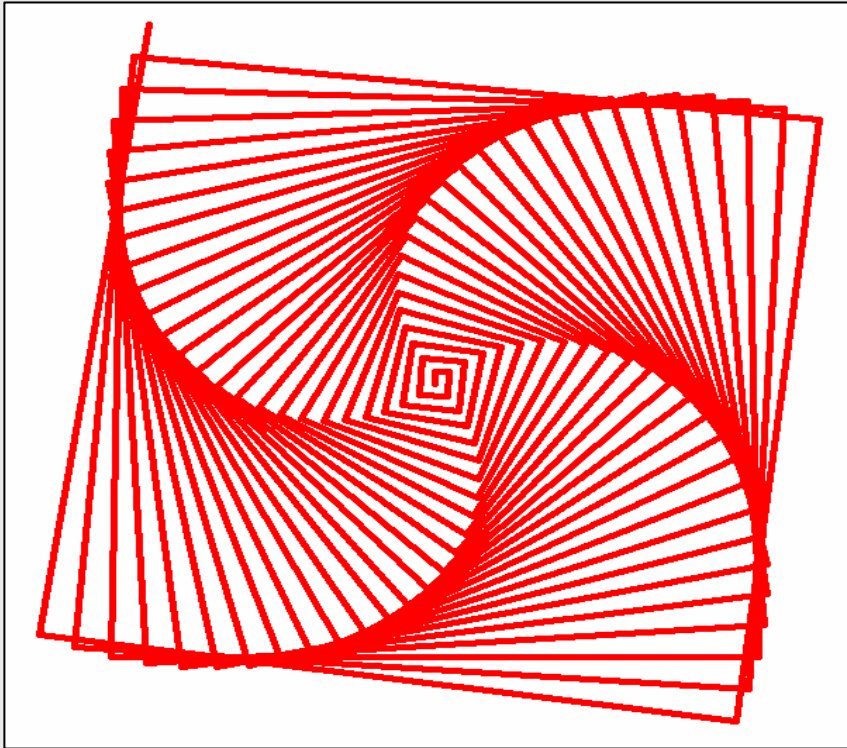
Saving your Programs

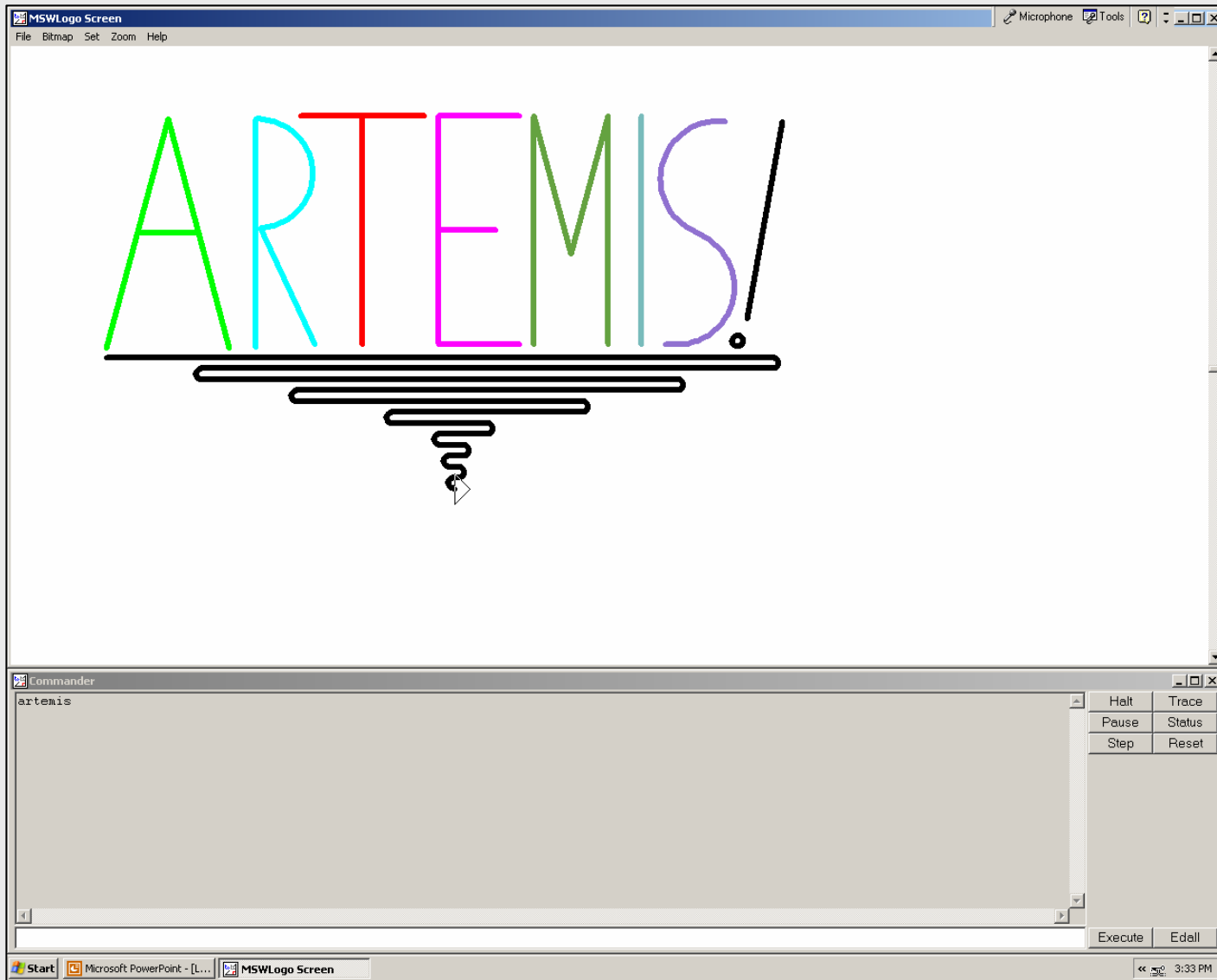
Since you will probably want to use the programs you wrote today some other time, you should **SAVE** them.

To do this, go to the “**FILE**” menu at the top of your screen and select “**SAVE**.”



Here's a few more interesting (and more complicated) things Logo can do:





Pretty
Nifty!

